

# T802 Study and Assessment Guide

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The Study and Assessment Guide is a key document from the point of view of knowing what you should be doing at any given point of the module. It will also direct you to other sources of information, library resources, the Preparing for T802 research website, and the documents which comprise the main module materials:

1. 'From synopsis to research proposal'
2. 'Literature review and referencing'
3. 'Methodology and techniques'
4. 'Theories and paradigms: their relevance in research'
5. 'Writing skills'
6. 'Data presentation, analysis and interpretation'
7. 'Research ethics'.

As is made clear in other parts of the module materials your study of T802 will not be a linear process. It will be more iterative in nature and you will need to find your own way through the multiple tasks which you will have to manage and complete. Of all the documents provided, this Study and Assessment Guide offers you the basis for navigating a course through your studies and managing it as a project. It takes you step by step through the five stages of the module, as shown in Table 1 of the *Welcome to T802* document, and indicates the work you have to submit in the four TMAs and the dissertation.

In conjunction with the Study planner, the Study and Assessment Guide also provides the framework for the work plan which you will produce during Stage 1 and which you will then use throughout the rest of the module.

The Study and Assessment Guide is therefore the backbone on which to build your studies. As well as reading it all when you start T802, we suggest that you also re-read it at the beginning of each stage of T802, to remind you of the key points for that stage, and also when preparing the assessment at the end of a stage, where it can be used as a checklist.

## Stage 1: Developing your research proposal

### 1.1 Introduction to Stage 1

Stage 1 will take you from your synopsis through to an approved proposal, which you will also submit as your first Tutor Marked Assignment (TMA 01). This stage may include any number of versions of your proposal between your first draft and the final version.

#### Study Note

This Study and Assessment Guide assumes that you have used the Preparing for T802 Research website to prepare and submit your synopsis prior to the start of the module. If you are fully registered and reading this from the module's presentation website but have not yet submitted your synopsis, then you should contact your supervisor, if allocated, or the Module Team urgently.

The research proposal stage is the first and possibly the most crucial stage of your research overall, which is why a substantial amount of study time is allocated to this stage. Its purpose is more than just the production of a written research proposal. It involves you developing both research skills and subject knowledge that you will be using throughout the research, culminating in the writing of your dissertation. Here is a list of the objectives of Stage 1 and the final expected outcome:

## **Objectives, outcome and actions for Stage 1**

### **Objectives**

During Stage 1 you will develop further aspects of your synopsis to include in your research proposal, specifically:

- elaborate your topic for T802 research
- identify further relevant and accessible sources of information for the research
- practice and develop further skills in accessing and reviewing relevant literature
- revise your research problem, and aim and objectives as needed
- plan how the research will be carried out
- schedule its activities
- develop writing skills in the production of a research proposal.

### **Outcome**

The outcome of Stage 1 is a complete and coherent research proposal that is a convincing description of what will be investigated, together with why, how and when it will be carried out. The material in the proposal forms the basis of the rest of your research. This outcome will be communicated in the form of your final research proposal and TMA 01.

### **Actions**

- Writing and submitting a first draft research proposal including:
  - Investigating further your research topic
  - Locating, accessing and reviewing further relevant literature
  - Identifying the practical problem more precisely
  - Justifying your intended research fully
  - Elaborating the research aim and objectives

- Selecting research methods and techniques (preliminary).
- Improving and iterating the research proposal including:
  - Refining your research aim and objectives
  - Selecting research methods and techniques (final)
  - Scheduling the activities
  - Assessing feasibility
  - Considering ethics
  - Assessing health and safety.
- Writing and submitting your final research proposal / TMA 01.

The key documents to study in Stage 1 are 'From synopsis to research proposal', 'Literature review and referencing' and 'Research ethics'.

The content of this guide for Stage 1 will help you develop your proposal, whose expected structure is summarised in Table 1 of this document.

### **Study note**

1. Past experience has shown that students approach this first stage of T802 differently: from having a clear idea of the intended area of study developed while preparing the synopsis, all the way to having almost no idea at all. The proposal phase is designed to cope with this variation and to bring everyone to more or less the same point for Stage 2.
2. If you are unclear about the direction of your research and perhaps found preparing the synopsis a challenge then it is important to make contact with your supervisor and seek guidance as early into the module as possible.
3. Instead, if you find the first draft proposal straightforward, perhaps because you already developed much of its content for your synopsis, then you are advised to submit it early and not wait for the initial proposal deadline (three weeks into the study calendar).

## **1.2 Topic, aim and objectives of your research**

You should go through Sections 2 to 4 of 'From synopsis to research proposal' carefully to ensure your research choices meet the requirements of a T802 project.

These will cover the following:

- What is expected from academic research and from you as a T802 student.
- What are appropriate choices of topic, research problem, aim and title for your project.

- What is meant by research objectives and how these are used to break-down your aim into actionable tasks.

### **1.3 Literature review, background and justification**

You should go through the 'Literature review and referencing' document carefully to remind you of the role of the literature review in your project and the key skills and techniques you should develop and apply.

Reviewing the literature will help you develop a good understanding of your chosen topic and current thinking in your field, and to help you frame and refine your research problem, so that you can establish the possible contribution to knowledge your research is going to make.

It will also allow you to explore the context of your work and justify why your research is needed.

### **1.4 Research design (methodology)**

During Stage 1, you should start thinking about how you will go about the research, even though you will lack a lot of the detailed knowledge you will acquire in later stages.

You should go through Section 5 of 'From synopsis to research proposal' carefully to learn about possible methodological choices for your T802 project.

In your choice, you should consider:

- Where and how will you carry out your own primary research?
- What kind of data will you require and how will you acquire it?
- What sort of processing or analysis will your data require?

### **1.5 Risk, feasibility and planning**

You should go through Section 6 of 'From synopsis to research proposal' to review how to manage project risk and assess feasibility. In addition, you should study Section 7 of that document to help you develop an initial work plan for your project.

It is essential that you consider the practicality of carrying out your research at this early stage. Research can easily flounder through lack of early attention to risk, including that arising from your methodological choices, and the resources you will need. It is not sufficient, for example, to have a vague intention of carrying out a survey, interviewing a number of experts, or experimenting in a laboratory without realistically assessing the availability of resources and the feasibility of obtaining enough relevant data.

You are expected to include a work plan as part of your research proposal in the form of a Gantt chart, showing your awareness of the main activities, milestones and deliverables of a T802 project, how they follow and depend on each other, and how they can be arranged within a reasonable schedule that takes into consideration your study time and other commitments.

### **1.6 Considering ethics**

Your research must be carried out to high ethical standards. Read the 'Research ethics' document to learn about the key ethical and legal issues which are pertinent to T802 research and may affect your project.

You will be required to consider such issues as part of your overall risk assessment, particularly in relation to potential human participants in your research. You will need to pay particular attention to health and safety, and the protection of personal data.

You will also be required to include a completed ethical checklist as part of your research proposal/TMA 01 submission. Such a checklist is included as an appendix of the proposal/TMA 01 template, which can be found under the Resources tab.

## **1.7 Writing your research proposal**

In preparing your research proposal, you are putting together a case with the purpose of persuading your supervisor that what you intend to do is appropriate and feasible. In addition, your research proposal should allow you to identify potential problems that might arise in the course of your research and take action against them or avoid them altogether. If you produce a vague proposal, then you are putting off decisions to a later date when they may cause severe problems. In broad terms, a good proposal will answer the following questions:

- What is the research problem that you will investigate?
- In what circumstances does it arise?
- What have other researchers discovered?
- What will you aim to achieve in the research?
- Why is your research relevant and justifiable?
- What methods will you use?
- What useful sources of information have you located?
- What are the principal tasks, deliverables and milestones involved?
- In what sequence will you carry them out?

The required contents of the research proposal are designed to help you to answer these questions. It is readily accepted that some of the detail may change as the work proceeds. However, because of time constraints, you need to be specific, practical and realistic in order to reduce the risk of failure as much as possible.

You need to write your proposal in appropriate language and using a sufficient level of detail for your intended audience – your supervisor and the T802 module team, markers and, by their proxy, the wider academic community. Your supervisor and members of the module team have considerable research experience appropriate to your qualification programme, so you do not need to explain basic principles and techniques. However, you should assume that your readership does not have specific knowledge in your particular area of investigation.

### **Study note**

Your final research proposal will be submitted as TMA 01 and will contain material that will form the basis of sections of TMA 02, though you will need to modify and extend it. Similarly, material will be carried from TMA 02 to TMA 03 and then to TMA 04, and from this to the dissertation. In this manner, information accumulates as the story unfolds and the writing task is made easier by retaining useful material rather than starting afresh for each document. However, it must be stressed that this is not just a cut and paste exercise.

Although the document ‘Writing Skills’ provides guidance which is useful primarily for the dissertation, it may have material relevant to the proposal and TMAs. In particular, it emphasises the need to ‘write up’ as you go along and develop your skills in this area over a period of time rather than leaving it all to Stage 5. You are strongly advised to at least skim read this document before submitting your first draft proposal.

### 1.7.1 Contents of the research proposal/TMA 01

For your research proposal you should use the template provided in the Resources section of the module website (under ‘Key documents’). All parts of the template must be completed in your final proposal/TMA01, including the ethical checklist in the appendix.

The main parts of the research proposal are summarised in Table 1.

**Table 1 The main parts of the T802 research proposal and related study materials in this guide**

| Part                                                                                                | Study materials                |
|-----------------------------------------------------------------------------------------------------|--------------------------------|
| Title                                                                                               | Section 1.2                    |
| 1. Introduction – including research background and justification                                   | Section 1.3                    |
| 2. Literature review – including both literature reviewed and planned review for Stage 2            | Section 1.3                    |
| 3. Research definition – including problem statement, aim and objectives and knowledge contribution | Section 1.2                    |
| 4. Research design (methodology) – including data, methods, and ethical and legal considerations    | Section 1.4<br><br>Section 1.6 |
| 5. Assessment of proposed research – including fit, resources and feasibility                       | Section 1.2<br><br>Section 1.5 |

## 1.8 Presentation of the research proposal

Your first draft proposal should be submitted to the discussion thread which you created in the appropriate T802 Proposals forum when you submitted your synopsis. Follow the instructions accompanying the forum to submit your first draft proposal and all subsequent ones.

**Do not put the proposal as text within the message or create a new discussion.**

Refer to the study planner on the module website for the due-dates for submission of the first draft and the final proposal as TMA01.

## 1.9 Improving and iterating the research proposal

Within a short time of submitting your first draft, you will receive feedback from your supervisor. Follow the advice given and produce a second version of the proposal. Initiate further discussion with your supervisor about any aspect of the feedback that isn't clear or to help with any queries that arise. Submit the revised proposal in the same way as the first. Continue in this way until your supervisor indicates that you have a proposal of acceptable standard. It is usual for at least three iterations to be required and, quite often, several more are needed. There is no limit to the number of iterations that can take place although, of course, they must be within the time period allotted for the proposal.

The word count of the final research proposal should be no more 4000 words, including all contents of the report except the list of references and the appendices. Your supervisor has been instructed to ignore any words over this limit.

## 1.10 Submitting TMA 01

Once your supervisor indicates that your proposal is acceptable, you should submit the final version as TMA 01 and continue with your research. Your assignment should be submitted using the online TMA/EMA system, and you should refer to its current regulations for any specific advice. There is no need for any interruption that the study planner might suggest if you are ahead of time with your final proposal. It is possible to lose momentum at this point. Most importantly, discuss with your supervisor if there are any areas of your proposal which, although acceptable, still need further attention.

## 1.11 If your proposal is not of acceptable standard

For most of the students who do not proceed beyond Stage 1 of T802 on a given presentation, it is a self-realisation process. They acknowledge that they are not ready to proceed and usually defer to a later presentation before the TMA 01 deadline after discussion with their supervisor.

For a small number of students who continue to engage with refining their proposal right up to the end of Stage 1, including the submission of TMA 01, the standard of one or more aspects may still fall short of what is required. In such cases, the proposal is regarded as *not of acceptable standard*, and this decision is based on your TMA 01 mark. If that happens, your supervisor will consult with the module team. Each

case is treated on an individual basis and feedback is given by the supervisor and a discussion on the best way forward usually follows.

Any student with a *not of acceptable standard* proposal at the end of Stage 1 has the right to carry on to Stage 2. However, the module team will recommend that serious consideration is given to deferring to a subsequent presentation, if that option is available. The majority of students who defer their studies at TMA 01 and do not go on to Stage 2 at the first attempt, return to T802 later and are successful in making progress.

## Study Note

Even if you do not have a final proposal of acceptable standard by the TMA01 due date, you must submit the latest version you have as TMA 01 by that date. In exceptional circumstances, your supervisor may be able to grant you a short extension.

## Study Note

- i. Students who are advised to defer by the T802 module team and who do not go on to Stage 2 are usually able to transfer 80% of their module fee to a subsequent presentation of T802 or any module (but see (ii–v) below). If you have an extension for TMA 01 then you may not receive feedback on it before the deadline to defer with this fee transfer. If you think you might wish to defer please contact your SST to discuss.
- ii. To be considered for 80% module fee credit, you must submit a TMA 01 which demonstrates your progress and work to date. A blank TMA 01 is not acceptable.
- iii. If you have deferred T802 once you cannot do so again.
- iv. There are regulations about how and when any transferred fee credit can be used and you will be advised about them when you ask your Student Support Team about deferral.
- v. Fees cannot be refunded, only transferred to a subsequent presentation.
- vi. If you continue to Stage 2 and TMA 02 against the recommendation of the T802 module team, you will not be able to access the 80% module fee credit if you decide to defer subsequently.

Further advice, in addition to that provided by your supervisor, may be sought from the Student Support Team (SST) – see your StudentHome page.

## Stage 2: Continuing the research

### 2.1 Introduction to Stage 2

This section covers the activities that culminate in the submission of TMA 02. Refer to the study planner for the due-dates for submission of this TMA. Stage 2 is about consolidating the work you have put into your proposal so that aspects will start to look like how they will in the dissertation. It will also be a time to revisit



parts of the module materials and study more intensely the parts that you skimmed earlier. This and subsequent sections will assume that, by now, you know your way around the various documents and specific links are not provided.

Here are the objectives of Stage 2, the final outcome (TMA 02) and the actions that you will need to carry out in order to achieve it.

## **Objectives, outcome and activities for Stage 2**

### **Objectives**

During Stage 2 you will further develop skills in:

- literature searching and reviewing
- identifying a suitable methodology
- planning how the research is to be carried out
- academic writing.

### **Outcome**

The outcome of Stage 2 is TMA 02, an interim report that comprises a detailed definition of the research problem or issue, including a review of the relevant literature, a preliminary consideration of the methodology to be used, and an update on the plan of activities.

### **Activities**

- Searching and reviewing the literature
- Developing the research definition
- Investigating methodology
- Preparing TMA 02
- Presentation of TMA 02.

## **2.2 Searching and reviewing the literature**

You have already begun the process of searching the literature and using it to help define your own primary research. You should now continue the process, looking deeper into the existing knowledge that will inform your own work. This should certainly be a major activity for Stage 2, continuing to a diminishing extent into Stage 3 and even into Stage 4. The literature review cannot be considered complete until you submit your dissertation.

During Stage 1 the emphasis was finding the gap in the existing knowledge to ensure a suitable niche for your study. In this second stage the emphasis will shift. As well as delving deeper into your chosen topic by finding and evaluating more sources, you will need to adopt a critical approach: to be analytical, not descriptive.

## 2.3 Developing the research definition

You will have made considerable progress in defining your research, having identified the problem or issue of concern, and achieved a growing understanding of it in the wider context that other researchers have explored. You know what you wish to achieve as an aim, so the big question is: what next? Your further progress with the literature search should put you in a stronger position to more clearly identify the data that you need in order to achieve your objectives. This is an extremely important element of your research, as it determines the research methods that you will use.

### Research questions

One frequently useful way to help define the data you need as you begin to define your research more closely is to formulate questions which arise from your objectives. These are often referred to as research questions as they help you define the direction of your research. It is a good idea to take a very systematic approach to thinking about this. One way to do this is as follows:

- Step 1: Using a three-column layout, list the objectives in the left-hand column.
- Step 2: Considering each objective in turn, in the middle column write down the questions that need to be answered in order to achieve the objective.
- Step 3: Considering each question in turn, in the final column note the various datasets that are needed to answer the question in full.

Here is a single example and this is taken further in the Additional resources.

Table 2 Example of a research question

| Objective                                                                                         | Question                                                                                                                                 | Data                                                      |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Identify, examine and discuss ERP driving forces, trends, and issues within the UK retail sector. | Q1. What is organisation type, sector, turnover, employee count, experience etc.? Used to check and classify organisation type and size. | D1. Statistics on the organisation in context.            |
|                                                                                                   | Q2. What are the forces driving ERP systems adoption for UK retail organisations?                                                        | D2. A ranked list of perceived driving forces.            |
|                                                                                                   | Q3. What are the perceived trends for ERP systems within the UK retail industry?                                                         | D3. A ranked list of trends and anticipated developments. |
|                                                                                                   | Q4. What are the key issues for ERP system selection for UK retail                                                                       | D4. A ranked list of perceived ERP system                 |

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## The iterative nature of research

When you first embark on your research, you may know relatively little about what is already known. Your objectives may well include finding out about aspects that turn out to be well covered by existing published material. Some of your objectives will therefore be met by the literature review. In some instances, it might be appropriate to amend the wording of the research aim – especially if, as the literature review progresses, it is clear that the original aim is largely met by secondary research. Similarly, there could be rewording of the objectives. The rewritten aim and objectives together with the research questions should form a fully coherent set. Your overall research aim is therefore met by a combination of primary research and secondary research.

This discussion serves as an illustration of the iterative nature of this aspect of your study at this stage. It is unlikely that your aim and objectives will be finally resolved until TMA 03 or even later.

The next step in ensuring that your research is coherent is to further clarify your research methods, tasks and deliverables. Your research methods (see Section 2.4) must be chosen clearly in response to the data required to achieve the objectives.

## 2.4 Investigating methodology

It is now time to return to the subject of methodology and consider in more detail what methods and techniques you will use. Reconsider what you wrote in your research proposal under 'Proposed research methods and techniques'. This aspect is covered in Section 3 of 'Methodology and techniques', so now is the time to study this document in its entirety if you have not already done so.

In your research proposal, under the heading 'Proposed research methods and techniques', you were required to describe the method(s) selected and justify them. You should now develop this further, and describe what data will be collected and how each selected method is likely to obtain the required data. Provide sufficient detail to make a convincing case. It is important that you clearly link the method to the data set required. One way to do this is to add a fourth column to Table 2 and record the research method(s) that you intend to use to acquire the data.

In due course, especially for later TMAs, you will need to justify your choice of research methods more thoroughly, but the above approach is a good starting point.

## 2.5 Preparing TMA 02

At this stage, having worked through the activities, you should be able to compile your TMA 02 using the contents in the following table. Do not omit any. Number the sections as indicated.

**Table 3 The content of TMA 02**

| Aspect | Action | Marks |
|--------|--------|-------|
|--------|--------|-------|

|                       |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
|-----------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Title of the research |                                                      | Check that it is still appropriate, or modify it.                                                                                                                                                                                                                                                                                                                                                                                                                                | 0  |
| 1                     | Introduction                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 25 |
|                       | 1.1 Background to the problem/issue                  | In this section, use the material from your research proposal but developed further as appropriate.                                                                                                                                                                                                                                                                                                                                                                              |    |
|                       | 1.2 Justification for the research                   | In this section, use the material from your research proposal but developed further as appropriate.                                                                                                                                                                                                                                                                                                                                                                              |    |
| 2                     | Research definition                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 35 |
|                       | 2.1 The practical problem/issue                      | As in the research proposal, but developed further in the light of your continued literature searching/reading.                                                                                                                                                                                                                                                                                                                                                                  |    |
|                       | 2.2 Existing relevant knowledge                      | This refers to the continuing process of reviewing the literature that you began when preparing your research proposal. The section has been retitled at this stage to reflect the broadening scope of your secondary research. You should now be looking at the wider context of the practical problem and be reviewing the literature in a critical manner.                                                                                                                    |    |
|                       | 2.3 Aim, objectives, methods, tasks and deliverables |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
| 3                     | Proposed methodology                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10 |
|                       | 3.1 Method(s) and techniques selected                | State the method(s) and techniques selected.                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |
|                       | 3.2 Justification                                    | Justify the use of the selected methods.                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
|                       | 3.3 Research procedures                              | Provide details of how you will collect and analyse the data. Show how the data collected will address any research questions that you have formulated and thereby allow the objectives to be achieved.                                                                                                                                                                                                                                                                          |    |
| 4                     | Planning and scheduling (updated)                    | Revisit your Gantt chart and update it as necessary.                                                                                                                                                                                                                                                                                                                                                                                                                             | 5  |
| 5                     | Progress to date                                     | Give a brief account of progress made so far. Include details of successful activities as well as problems encountered. In the latter case, outline what steps are being taken to reduce or avoid the problems. Refer to anything which has changed but which is not otherwise included in the TMA. For example, you may need to update the risk and ethics section of the proposal in the light of changes in methodology. In such cases use an appendix to record the changes. | 5  |

|              |                                                                        |    |
|--------------|------------------------------------------------------------------------|----|
| References   | Update your reference list and check its accuracy.                     | 10 |
| Presentation | Check:                                                                 | 10 |
|              | the clarity and accuracy of written English, and of figures and tables |    |
|              | the use of correct spelling, grammar and punctuation                   |    |
|              | overall coherence including the structure and organisation of material |    |
|              | the use of a clear and readable font, such as Times New Roman, 11 pt.  |    |

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## 2.6 Presentation of TMA 02

Your assignment should be submitted using the online TMA/EMA system, and you should refer to its current regulations for any specific advice. Refer to the study planner for the due dates for TMA 02 submission.

The word count should be no more than 4000 words. Any words over this will not be considered. Figure captions and tables are included in the word count. The word count excludes the list of references, appendices (see Section 5.6) and the preliminary pages (see below). You should not try to reduce the word count by inappropriate use of appendices or the inclusion of text in a format (such as pdf or image) which is excluded from software word counts. If you have a text heavy figure that you need to include, then check with your supervisor that your use of the medium is appropriate before submitting it.

Provide a caption for each figure and table, and number them according to the convention in Section 5.7. Ensure that you refer to each of these in the text.

Include the following preliminary sections.

1. A title page comprising:

- T802 Research project
- MSc in title of your intended MSc, as appropriate
- Your name
- Student personal identifier
- Date
- Title of the research.

2. A contents page.

## Stage 3: Progressing the research

## 3.1 Introduction to Stage 3

This section covers the activities that culminate in the submission of TMA 03. Consult the study planner for the due date for this submission.

Here are the objectives of Stage 3, the final outcome and the actions that you will need to carry out in order to achieve it:

### Objectives, outcome and activities for Stage 3

#### Objectives

During Stage 3 you will:

- develop skills in identifying a suitable methodology and planning how the research is to be carried out
- begin the process of data collection
- further develop writing skills in producing TMA 03.

#### Outcome

The principal outcome of Stage 3 is TMA 03, a report that comprises a detailed definition of the research problem or issue (as per TMA 02 with appropriate modifications or additions), an explanation of the methodology and the detailed methods to be used, an account of preliminary data collection (including the testing of methods or techniques) and an update on the plan of activities.

#### Activities

- Reviewing TMA 02
- Designing the methodology and methods
- Planning the data collection
- Preparing TMA 03
- Presentation of TMA 03.

## 3.2 Reviewing TMA 02

Your first activity of Stage 3 is to consider carefully the feedback you have received for TMA 02. Use it as a trigger for reflection and to review such matters as time management and progress to date. Discuss the feedback with your supervisor, particularly in any cases where you are unsure how to address the feedback. It is important early in Stage 3 to consider how you can make improvements that can be incorporated into TMA 03.

### 3.3 Designing the methodology and methods

At this stage you need to give detailed consideration to the methods and techniques that you plan to use as you prepare to begin primary data collection. Reconsider what you wrote in TMA 02 and use 'Methodology and techniques' as a resource as necessary. You should be able to add more detail about the techniques. Remember that, in particular, for TMA 03 you need to be able to demonstrate how they will lead to acquisition of the data that will be needed to achieve your objectives. You should include appropriate data analysis techniques – it is one thing to collect a large amount of quantitative or qualitative data, but quite another to turn that into relevant information.

### 3.4 Planning the data collection

Having reached the stage where you have a good understanding of the problem or issue being researched and having selected appropriate methods and techniques, you are now in a good position to plan the necessary data collection in detail.

For the purposes of TMA 03, you need to be able to identify the method(s) you have chosen and describe the data-collection technique(s) selected. Justify your choices not only by explaining how they will provide the required data but also by demonstrating the practicability of using them.

You should now be taking some practical steps towards data collection, whether it is assembling apparatus, contacting potential interviewees, designing and testing a questionnaire, gaining permission to visit an organisation or searching for essential documents. You may even achieve some real data collection at this stage as a result of these preliminary steps.

Bear in mind that you will ultimately need to provide a considerable amount of detail about your research methods – this should be such that another researcher could repeat your research to achieve the same results. Much of the detailed information (such as the questions in a questionnaire) can be placed in appendices.

### 3.5 Preparing TMA 03

Having worked through the Stage 3 activities, you should be able to compile TMA 03 using the contents in the following table. Number the sections as indicated. As you will use material which you have carried over from TMA 02, it is important to clearly indicate which text is unchanged and which is new. One way to do this is to use a different colour for the revised text. It is worth discussing this with your supervisor.

**Table 4 Contents of TMA 03**

| Aspect                              | Action                                                                              | Marks |
|-------------------------------------|-------------------------------------------------------------------------------------|-------|
| Title of the research               | Check that it is still appropriate, or modify it.                                   |       |
| 1 Introduction                      |                                                                                     | 10    |
| 1.1 Background to the problem/issue | In this section, use the material from TMA 02 but developed further as appropriate. |       |

|   |                                                      |                                                                                                                                                                                                                                                                                                                                                          |    |
|---|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
|   | 1.2 Justification for the research                   | In this section, use the material from TMA 02 but developed further as appropriate.                                                                                                                                                                                                                                                                      |    |
| 2 | Research definition                                  |                                                                                                                                                                                                                                                                                                                                                          | 20 |
|   | 2.1 The practical problem/issue                      | As in TMA 02, but developed further in the light of your continued literature searching/reading.                                                                                                                                                                                                                                                         |    |
|   | 2.2 Existing relevant knowledge                      | This refers to the continuing process of reviewing the literature that you began when preparing your research proposal and developed further for TMA 02.                                                                                                                                                                                                 |    |
|   | 2.3 Aim, objectives, methods, tasks and deliverables | As for TMA 02, with any refinements needed.                                                                                                                                                                                                                                                                                                              |    |
| 3 | Proposed methodology                                 |                                                                                                                                                                                                                                                                                                                                                          | 40 |
|   | 3.1 Method(s) and techniques selected                | State the method(s) selected.                                                                                                                                                                                                                                                                                                                            |    |
|   | 3.2 Justification                                    | Justify the use of the selected methods.                                                                                                                                                                                                                                                                                                                 |    |
|   | 3.3 Research procedures                              | Provide details of how you will collect and analyse the data, and how the data will enable answers to the research questions and/or address the objectives. Include a description of any 'pilot' activities, such as trial questionnaires or test analyses of results.                                                                                   |    |
| 4 | Planning and scheduling (updated)                    | Revisit your Gantt chart and update it as necessary.                                                                                                                                                                                                                                                                                                     | 5  |
| 5 | Progress to date                                     | Give a brief account of progress made so far. Include details of successful activities as well as problems encountered. In the latter case, outline what steps are being taken to reduce or avoid the problems. Refer to anything which has changed but which is not otherwise included in the TMA. In such cases use an appendix to record the changes. | 5  |
|   | References                                           | Update your reference list and check its accuracy.                                                                                                                                                                                                                                                                                                       | 10 |



- the clarity and accuracy of written English, and of figures and tables
  - the use of correct spelling, grammar and punctuation
  - overall coherence including the structure and organisation of material
  - the use of a clear and readable font, such as Times New Roman, 11 pt.
- 

## 3.6 Presentation of TMA 03

Your assignment should be submitted using the online TMA/EMA system and you should refer to its current regulations for any specific advice. Refer to the study planner for the due-dates for TMA 03 submission.

The word count for TMA 03 should be no more 6000 words. Any words over this will not be considered. Figure captions and tables are included in the word count. The word count excludes the list of references, appendices (see Section 5.6) and the preliminary pages (see below). You should not try to reduce the word count by inappropriate use of appendices or the inclusion of text in a format (such as pdf or image) which is excluded from software word counts. If you have a text heavy diagram that you need to include, then check with your supervisor that your use of the medium is appropriate before submitting it.

Provide a caption for each figure and table, and number them according to the convention in Section 5.7. Ensure that you refer to each item in the text.

Include the following preliminary sections.

1. A title page comprising:

- T802 Research project
- MSc in title of your intended MSc, as appropriate
- Your name
- Student personal identifier
- Date
- Title of the research

2. A contents page.

## Stage 4: Further progressing the research

### 4.1 Introduction to Stage 4

This section covers the activities that culminate in the submission of TMA 04. Consult the study planner for the submission date.

Here are the objectives of Stage 4, the final outcome and the activities that you will need to carry out in order to achieve it.

## **Objectives, outcome and actions for Stage 4**

### **Objectives**

During Stage 4 you will:

- collect the data, as planned in the Stage 3 methodology
- analyse and/or process the data for purposes of presentation and comprehension
- evaluate the data in the context of the objectives
- further develop writing skills in producing a fourth TMA.

### **Outcome**

The principal outcome of Stage 4 is TMA 04, which is developed from TMA 03 but (additionally) describes the data collected, its analysis and evaluation.

### **Actions**

- Reviewing TMA 03
- Collecting the data
- Analysing, representing and presenting the data
- Evaluating the data
- Preparing TMA 04
- Presentation of TMA 04.

## **4.2 Reviewing TMA 03**

Your first activity is to consider carefully the feedback you have received for TMA 03. Use it as a trigger for reflection and to review such matters as time management and progress to date. Discuss the feedback with your supervisor, particularly in any cases where you are unsure how to address the feedback. It is important early in Stage 4 to consider how you can make improvements that can be incorporated into TMA 04.

## **4.3 Collecting the data**

Throughout Stage 4 you will be accumulating data and information from various sources, such as interviews, experiments, surveys and documents. Make sure that you are sufficiently well-organised in recording and indexing everything you obtain. Keep related data together and note what it is and when it was collected, plus other relevant details.

It is likely that the data that you collect will be both in physical and electronic form. Photocopy or scan vital physical records of data if there is any possibility of losing it. Don't forget to make regular and frequent backups of any electronically stored data, your draft TMAs and the dissertation. If you are following the advice given in 'Preparing for research' to keep a research diary, then ensure that it is kept up to date.

## **4.4 Analysing, representing and presenting the data**

Having collected the data, whether quantitative or qualitative, attention must be given to how it is processed. There are three reasons for this:

- to check that all the data needed has been acquired
- to turn it into meaningful information
- to communicate it effectively in the TMA and the dissertation.

At this stage, if you have not already done so, you should study the document 'Data presentation, analysis and interpretation'. There are also more sources of information on this topic in the 'Additional reading' document in the Additional resources.

If you collect large quantities of data, you need to assess from time to time what you have obtained. A clear overview should then make it easier to spot gaps and identify other aspects to search for. You are no doubt aware of the various techniques for condensing and summarising quantitative data, including finding averages, medians and spreads, and representing data graphically in line graphs, histograms (bar charts) and pie charts. Compiling these as you proceed can be a useful means of keeping track of what you have obtained. Qualitative data can also be summarised as you go along – for example, using tables to record outline responses to the various questions asked. Using statistics is a useful source of information on working with statistical data.

You may be using specific analytical techniques to make sense of your data – for example, using mass balance calculations as part of a waste management study, or carrying out content analysis of interview responses to identify key issues. Now is the time to check that these techniques 'work' with your raw data and yield information relevant to your objectives. Your TMA 04 should include examples of any analysis of this sort.

## **4.5 Evaluating the data**

At this stage you are mainly concerned with the collection and representation of data, including appropriate analysis. However, inevitably (and sensibly) you will be looking critically at what you have obtained in the context of achieving your objectives. It is important to know at this stage whether the data collected allows you to achieve your objectives in full. If not, is additional data acquisition worthwhile and feasible? Answering these questions allows you to adjust your approach 'in real time' and not waste a lot of time in duplicating work or getting off-track.

Whilst you should not attempt to reach conclusions about your research prematurely, you may have some tentative conclusions that can be referred to in your TMA 04 and perhaps discussed with your supervisor.

## 4.6 Preparing TMA 04

At this stage, having worked through the activities, you should be able to compile TMA 04 using the contents of the following table. Number the sections as indicated. As with TMAs 02 and 03, reuse material where it is still valid; otherwise adapt and develop it further. As you will use material which you have carried forward from TMA 03 it is important to clearly indicate which text is unchanged and which is new. One way to do this is to use a different colour for the revised text.

You will notice that an abstract is required at the beginning of TMA 04, immediately after the title. This has been introduced because it is a feature that will be required in the dissertation. The abstract is a brief summary of the whole story of your research, as far as it has progressed to date, so clearly you will have more to add later. Write, in around 200–300 words, the issue or problem researched, how and where it arises, what is its impact, what you aim to achieve, the methods and techniques you are using, and any results obtained so far. Do not include any headings, figures, tables and use bullet points with care. Write in continuous text using paragraph breaks to delineate different parts. It is not usual to cite any references within the abstract, except in cases where an aspect of the work is based on or closely associated with a previous study.

**Table 5 Contents of TMA 04**

| Aspect                       | Action                                            | Marks                                                                                            |
|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Title of the research</b> | Check that it is still appropriate, or modify it. |                                                                                                  |
|                              | <b>Abstract</b>                                   | 10                                                                                               |
| <b>1</b>                     | <b>Introduction</b>                               | 5                                                                                                |
|                              | <b>1.1 Background to the problem/issue</b>        | In this section, use material from TMA 03 but developed further as appropriate.                  |
|                              | <b>1.2 Justification for the research</b>         | In this section, use material from TMA 03 but developed further as appropriate.                  |
| <b>2</b>                     | <b>Research definition</b>                        | 15                                                                                               |
|                              | <b>2.1 The practical problem/issue</b>            | As in TMA 03, but developed further in the light of your continued literature searching/reading. |
|                              | <b>2.2 Existing relevant knowledge</b>            | As in TMA 03, but developed further in the light of your continued literature searching/reading. |
|                              | <b>2.3 Aim, objectives,</b>                       | As in TMA 03, developed further as appropriate.                                                  |

|          | methods, tasks<br>and deliverables                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3</b> | <b>Methodology</b>                                                                                                                                                                                                                                                                                                                                    | <b>15</b>                                                                                                                                                                                                                                                                                                                                                |
|          | <b>3.1 Method(s) and techniques selected</b>                                                                                                                                                                                                                                                                                                          | As in TMA 03, developed further as appropriate.                                                                                                                                                                                                                                                                                                          |
|          | <b>3.2 Justification</b>                                                                                                                                                                                                                                                                                                                              | As in TMA 03, developed further as appropriate.                                                                                                                                                                                                                                                                                                          |
|          | <b>3.3 Research procedures</b>                                                                                                                                                                                                                                                                                                                        | As in TMA 03, developed further as appropriate.                                                                                                                                                                                                                                                                                                          |
| <b>4</b> | <b>Results</b>                                                                                                                                                                                                                                                                                                                                        | <b>30</b>                                                                                                                                                                                                                                                                                                                                                |
|          | <b>4.1 Data collected</b>                                                                                                                                                                                                                                                                                                                             | Report the results of your research to date, including the 'raw' data collected and the results of any subsequent analysis or processing.                                                                                                                                                                                                                |
|          | <b>4.2 Analysis</b>                                                                                                                                                                                                                                                                                                                                   | Report the results of your research to date, including the 'raw' data collected and the results of any subsequent analysis or processing.                                                                                                                                                                                                                |
| <b>5</b> | <b>Planning and scheduling (updated)</b>                                                                                                                                                                                                                                                                                                              | <b>5</b>                                                                                                                                                                                                                                                                                                                                                 |
| <b>6</b> | <b>Progress to date</b>                                                                                                                                                                                                                                                                                                                               | <b>5</b>                                                                                                                                                                                                                                                                                                                                                 |
|          |                                                                                                                                                                                                                                                                                                                                                       | Give a brief account of progress made so far. Include details of successful activities as well as problems encountered. In the latter case, outline what steps are being taken to reduce or avoid the problems. Refer to anything which has changed but which is not otherwise included in the TMA. In such cases use an appendix to record the changes. |
|          | <b>References</b>                                                                                                                                                                                                                                                                                                                                     | <b>5</b>                                                                                                                                                                                                                                                                                                                                                 |
|          | <b>Presentation</b>                                                                                                                                                                                                                                                                                                                                   | <b>10</b>                                                                                                                                                                                                                                                                                                                                                |
|          | Check:                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                          |
|          | <ul style="list-style-type: none"> <li>the clarity and accuracy of written English, and of figures and tables</li> <li>the use of correct spelling, grammar and punctuation</li> <li>overall coherence including the structure and organisation of material</li> <li>the use of a clear and readable font, such as Times New Roman, 11 pt.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                          |

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## 4.7 Presentation of TMA 04

Your assignment should be submitted using the eTMA system and you should refer to its current regulations for any specific advice. Refer to the study planner for the due dates for TMA 04 submission.

The word count should be no more than 9000 words. Any words over this will not be considered. Figure captions and tables are included in the word count. The word count excludes the list of references, appendices (see Section 5.6) and the preliminary pages, see below. You should not try to reduce the word count by inappropriate use of appendices or the inclusion of text in a format (such as pdf or image) which is excluded from software word counts. If you have a text heavy figure that you need to include, then check with your supervisor that your use of the medium is appropriate before submitting it.

Provide a caption for each figure and table and number them according to the convention in Section 5.7. Ensure that you refer to each item in the text.

Include the following preliminary sections:

1. A title page comprising:

- T802 Research project
- MSc in title of your intended MSc, as appropriate
- Your name
- Student personal identifier
- Date
- Title of the research.

2. A contents page.

There is a final, practical concern for TMA 04. If you intend to import material from other software into your dissertation, then it is worth using TMA 04 as a test of whether it survives the submission process. Use an appendix to TMA 04 for this purpose.

## **Stage 5: Completing the research and dissertation**

### **5.1 Introduction to Stage 5**

The dissertation tells the story in a manner appropriate to your audience: namely, your examiners. It should be a clear, straightforward, interesting account, in plain English, and focused on a significant piece of research.

Having said what a dissertation is, it is perhaps worth adding that it is not:

- a company report
- a project report
- a consultant's report

- a review of current information
- an instruction manual.

Writing the dissertation should have been made easier through the preceding process of evolution; it is not intended to be a final, formidable task to be worked on in isolation. Throughout the period of research, from first formulation of the proposal, written material has been added bit by bit until the final dissertation emerges. Your research proposal has evolved through the various stages and their associated TMAs. At Stage 5 the final results and conclusions are to be added. In this manner, information has been collected as the story unfolds and the task made easier. Although significant editing and additional material are needed at this final stage, the principle is to retain useful material and then add to it until the dissertation is complete.

Here are the objectives of Stage 5, the final outcome and the activities that you will need to carry out in order to achieve it.

## **Objectives, outcome and actions for Stage 5**

### **Objectives**

During Stage 5 you will:

- complete the data collection
- complete the data analysis and interpretation
- reach conclusions achieved through the research
- further develop skills in writing the dissertation.

### **Outcome**

The outcome of Stage 5 is the completed dissertation that describes the research carried out, including its origins, its context, the methods used, the results obtained and the overall conclusions reached.

### **Actions**

- Reviewing TMA 04
- Finishing your research
- Drawing conclusions
- Completing the dissertation
- Adding the extended abstract
- Structuring the dissertation
- Formatting and presentation

- Finishing the job
- Submitting the dissertation.

## 5.2 Reviewing TMA 04

Your first activity is to consider carefully the feedback you have received for TMA 04. Discuss this with your supervisor, particularly in any cases where you disagree, and consider how you can make improvements to be incorporated within the dissertation.

## 5.3 Finishing your research

The end is in sight, but there is still some significant work to do. You should not underestimate the amount of time required to tie up all the loose ends, complete the data collection, write Chapters 4 and 5, and finally tidy up the whole dissertation and submit it for assessment. Typically, this final stage occupies the last two months of the module so it is worth revisiting your work schedule for reassurance that what you have still to complete can be fitted within the final weeks. Decide which interviews, tests or visits you really must complete and plan when you will write the final chapters and finalise the abstract. It is dangerous to let too much time pass – if you do, you could end up in the situation where you suddenly realise that you have one week to go before the deadline and too much work to complete.

## 5.4 Drawing conclusions

At this late stage, there is a danger of being overwhelmed by a mass of ideas that derive from all the information you have painstakingly collected, and losing sight of what you are trying to achieve. Don't panic! Consider the process of drawing conclusions in six steps:

1. Reach conclusions about the detailed data you obtained from your experiment, case study or whatever method(s) you used. This is mainly a question of organising and presenting them in a form that enables detailed evaluation.
2. Use the conclusions to indicate the extent to which you have achieved your objectives.
3. Consider how (and if) you have satisfied the overall aim of your research.
4. Stand back to consider the wider implications of your conclusions: in what way are they significant to situations other than the one that you observed?
5. Consider how your conclusions relate back to the literature review.
6. Reflect on the research process you have undertaken and what you have learned from it. How could what you have learned benefit any future research activity that you might be involved in?

## 5.5 Compiling the dissertation

At this stage it is worth returning to 'Writing Skills' to remind you of a number of aspects to keep in mind while completing the dissertation. In particular, reread Section 3 on structuring the dissertation and Section 4 on writing style and technicalities. Again, I emphasise that the dissertation should be written to tell the story of your research, although it should not be a 'blow by blow' account of what happened during the entire period. It should be much more than a sequential collection of facts, observations and discussion.



## A note about use of the correct tense

In telling the story, you will be using much of the material you wrote for previous TMAs. Recycling all your carefully worked-out explanations will save a lot of work. However, don't forget that time has moved on. Where you previously described what will happen, in such language as 'the stated aim will be pursued' and 'the required data will be collected using ...', you are now writing an account of what has happened. Therefore, you need to edit all recycled material carefully using the appropriate past tense: 'the stated aim was pursued' and 'the required data was collected using ...'. See the document 'Verb Tenses in Academic Writing' for further discussion of the use of tenses in report writing.

## 5.6 Structuring the dissertation

The advised structure for the dissertation is given in the Dissertation template in Assessment.

### Notes about the structure

Whilst it is highly advisable to follow the template structure as closely as possible (as the marking scheme is closely linked to it) there may be very good reason to modify it in your particular circumstances. If, after reading the rest of this section, you consider that you may want to deviate from it in a significant way, then discuss this with your supervisor. One important aspect of the template to follow is the format of preliminaries and the division of chapters. So, do not omit any of the main sections listed, except for very good reason. There are, however, some aspects where it may be desirable to exercise some flexibility:

- Adapt any sub-headings according to your specific research.
- If you feel the need for introductions to chapters, then either you can start the chapter with a short unnumbered section – that is, after the title and before Section X.1 – or, for a longer introduction, you can call it Section X.1, as in '3.1 Introduction to ...'.
- Don't forget that in long sections you can help to clarify your story by dividing the text into subsections. You can choose to number these (in which case include them in the contents list), or provide expressive headings. Without these signposts it is otherwise very daunting for a reader to face many pages of continuous text with little indication of the 'concerns' being described. When numbering sections, don't go below the third level, for example 1.2.3. This topic is further discussed in Sections 3.2 and 3.3 of 'Writing Skills'.

### Writing the abstract

You have already encountered the abstract in TMA 04 and should now develop it further. It is likely to be one of the last, if not the last, sections you complete, even though it is the first item in the dissertation after the title page. The abstract is not an introduction to the dissertation. It must tell the whole story of your research and cannot therefore be written until the dissertation is (at least nearly) completed. The abstract will typically be read as a stand-alone item; that is, it should be understood without reference to any other part of the dissertation. Indeed, a reader who wanted to search a number of such dissertations for information on a particular issue would start by reading the abstracts only, rather than spending a lot of time reading long sections of text.

In 'telling the whole story' the abstract needs to explain in brief what the issue or problem researched was, how and where it arises, what its impact is, what you aimed to achieve, the methods and techniques you

used, the results you obtained and the conclusions you reached. Clearly, in the space of around 250–300 words you are only going to outline each of these. However, you need to sell your research and capture the interest of your reader in the short time it takes to read your abstract.

## Notes

- The abstract should be written as continuous text, separated into paragraphs. Use lists and bullet points sparingly.
- Write it as a description of work that has been carried out, not work that will be carried out.
- It should contain 250–300 words printed on a single page.
- Do not include any figures, tables, drawings or photographs.
- It is not usual to cite any references within the abstract, except in cases where an aspect of the work is based on or closely associated with a previous study.

## Writing the chapters

### Chapter 1 Introduction

Sections 1.1–1.2 are likely to be derived from what you wrote in TMA 04.

Section 1.3 can be used to explain any terminology that is not widely understood or that is perhaps controversial and subject to different interpretations.

Section 1.4 is used to describe the scope or extent of the research so that it is clear what has been investigated and what has not. This might apply to particular organisations, roles, processes, products, regions and timescales.

Section 1.5 should be a brief description of the contents of the dissertation, chapter by chapter.

### Chapter 2 Research definition

This is likely to be based on what you wrote in TMA 04. Do check that you have included your latest secondary research and cite any references correctly.

### Chapter 3 Methodology

Sections 3.1 and 3.2 will probably also be similar to what you wrote in TMA 04, but now with additions resulting from your experience in carrying out the research.

Similarly, in Section 3.3 you can now describe the research procedures as they were carried out. Here you will show how you have carried out your research techniques according to accepted practice and account for any variations to this. It is also appropriate to acknowledge any constraints on your research imposed by the T802 context. If these exist it is likely that you will also refer to them in Sections 4.2 and 5.4.

In Section 3.4 describe any ethical considerations, even if you did not encounter any particular difficulties.

## Chapter 4 Analysis and interpretation

You may wish to adapt the chapter structure in order to tell your version of 'analysis and interpretation'. Whilst you still need to maintain the logical sequence of describing the data collected, how it was analysed and how it was interpreted in relation to your objectives, you can modify the chapter structure to handle complexity, for example if you used multiple case studies. However, you organise the early part of this section, there should be a separate section (Section 4.3 in the template) which deals with each objective in turn. Similarly, there should be a separate section (Section 4.4 of the template) which includes explicit reference to the output that results from achieving the aim. For example, if the aim was to produce a set of recommendations, then they should be listed here.

## Chapter 5 Conclusions

This is likely to be the shortest chapter but is perhaps the most crucial. It should round off the story by bringing together the interpretations from the previous chapter, summarising them and offering concluding remarks. The purpose is to demonstrate to the reader the extent to which the aim and objectives have been achieved.

Do not bring in new data or new considerations in Sections 5.1 and 5.2.

In Section 5.3 suggest how the work might be continued, by you or by other interested researchers.

In Section 5.4 you have the opportunity to round off the dissertation with some conclusions about the broader context of the research, beyond the research aim you pursued. You may wish to use the suggested heading 'Implications of the research', or you can amend it or introduce more than one section or 'concern'. Aspects you might wish to address include:

- how your results could be significant in other situations
- the implication of your results in other contexts, e.g. legal, financial, political, social
- whether your conclusions have further ethical implications
- how implementation of your conclusions might be relevant to other problems or issues
- the effectiveness of the research methods you employed
- the limitations on your results and conclusions arising from, say, the constraints of your research in the T802 context.

In Section 5.5 you are asked to reflect on the T802 experience and what you have learned. What, for example, would you do differently if you were to start again? How has what you have done for T802 helped to prepare you for any future research you may carry out? What has been the significance for your personal development of carrying out the research?

## Use of summaries

It is often a very effective end to a particular chapter to include a summary. By the end of the module, as you write the dissertation you will be very close to all the material and will not possibly be able to understand how anyone can fail to understand the arguments that you are putting forward. You need to

remember, however, that anyone reading the dissertation will not have lived with it for the best part of 12 months! So a summary can highlight important aspects of the chapter that you need the reader to carry with them as they read on.

## References

List these using the OU Harvard method, as described in Section 5 of Academic writing, plagiarism and referencing – references and citations. You should check references for accuracy, and check that there is an exact correspondence between citations in the chapters and the references listed. Do not add a bibliography or section of further or recommended reading.

## The extended abstract

The purpose of an extended abstract is to allow you to give a synopsis of your research, targeted at a more general audience than is the case for the main dissertation.

Note that the extended abstract is in addition to the abstract that appears at the beginning of your dissertation. The extended abstract is not just a longer version of the normal abstract: it is written to a more general audience and the restrictions on formatting and citations do not apply. Hence it is appropriate for the extended abstract to contain visual aids to summarise the work, such as a table or diagram. In addition, citing key papers which support your work is also appropriate.

## Content of the extended abstract

The extended abstract should summarise the key points in the dissertation and be written at a level appropriate for an audience which knows something about your subject but who are not experts. It should be easily read and understood by someone with only a superficial knowledge of your subject.

It should include:

- a summary of the motivation and research aim (or equivalent)
- a summary of the methodology
- key results
- a summary analysis
- a summary discussion
- key outcomes.

An extended abstract will, typically, not include elements of a literature review and will only contain key citations and references. It is possible for an extended abstract to contain no references.

Trying to create the extended abstract by simply copying and pasting text from your dissertation rarely works. Because the extended abstract and the main dissertation are aimed at different audiences, you will probably find that you need to simplify some of the technical language, and provide more structure to allow the extended abstract to flow.

## Format of the extended abstract

The extended abstract should be submitted as the first appendix to your dissertation. It should be possible for it to stand alone if extracted from the dissertation, so the Appendix title page should precede it and if you have a running header or footer it should not appear on the extended abstract pages. The word limit is 1250. It will typically contain about 2/3 pages of text and some number of pictures/figures/tables/charts in addition to the text. Overlong extended abstracts are unlikely to receive the highest mark and in all cases only the first 6 pages will be marked.

You should use the same font and layout as for the rest of your dissertation. Although you need not number non text items, it should be clear to which item you are referring (e.g. the figure below shows ...). An example extended abstract is given in the Additional resources.

## Appendices

Additional material is often placed in one or more appendices to provide detailed evidence in support of the summary information in the main body of your dissertation. The appendices will not be marked in the same way as the body of the dissertation but will be checked for consistency with it. They may also be useful to help someone taking the research further. Typical examples might be tabulated results from experiments, interview notes or transcripts, and the detail of surveys that might otherwise take up too much space in Chapters 3 and/or 4.

Ensure that all appendices are directly referred to somewhere in the text. Give each appendix a title.

Only include material in an appendix if it is strictly relevant to your narrative. Do not use appendices to add bits of vaguely relevant material, such as promotional documents, biographies, lengthy descriptions of processes, technical specifications, contracts and legal papers.

## An important note about 'success'

In carrying out research, we all strive for and anticipate success. However, in T802 – and indeed in any technological research – it is rare to find a solution to the big problem that has eluded researchers for many years. You may well achieve some progress in a certain area but not in another, leaving scope for further work by you, or by others. The hypothesis you set out to support may turn out to be flawed. The experiment you carried out may have produced unexpected (and unwelcome) results. However, when you hit an unexpected problem, you should try to learn something from it, to be in a better position to make progress via another route. 'Success' can therefore be understood in terms of learning and making progress, rather than finding an answer and drawing a neat line under it. Similarly, 'failure' is not about failing to find an answer, but more about not learning from experience.

In the context of T802, you should describe precisely what you observed and discovered. There is no need to hide unwelcome or contradictory results; they won't lose you marks. Write your conclusions critically and try to explain any unexpected results. You are bound to have some. Indeed, if everything in a piece of research went according to plan and produced the expected answer, one is entitled to ask if it was worth carrying out in the first place.

## 5.7 Formatting and presentation

The standards for layout and presentation are described below. It will save you a lot of time to use the standard template. However, if you don't, please follow the instructions carefully. You are not expected to

meet the detailed standards used in professional technical publications, but you should aim to achieve a clear and consistent layout.

The module Examination and Award Board can refuse to examine work that is incorrectly presented.

## Length of dissertation

The length must be between the minimum of 10 000 words and the maximum of 15 000 words. This applies to Chapters 1–5 only, and excludes the abstract, preliminary pages, reference list and appendices. Advice about what is included (and what is not included) in the word count is the same as for the TMAs.

The maximum word count must be taken seriously. Dissertations longer than 15 000 words may not be examined.

## Fonts, typography and layout

Always keep in mind the need for your readers to be able to read your work without undue strain. Although the dissertation is submitted electronically there are certain basic formatting points to remember:

- Use a standard font that is easy to read, e.g. Times New Roman, Arial. It is best to avoid Microsoft specific fonts, such as Calibri, since these may get converted when they are read using different software.
- Choose a font size that is easy to read but doesn't waste space – 11 or 12 point is acceptable. 11 point is recommended if you are using Times New Roman.
- Leave sufficient margins on both the left and the right of the page, approximately 2 cm either side. This corresponds to 'moderate' margin settings on MS Word.
- Use 1.5 line spacing.
- Avoid over-fussy, multiple use of word-processing facilities, such as mixing too many font types and point sizes, or overuse of combinations of emboldened, italicised, coloured and underlined text.

## Pagination

Pages must be numbered, including references and appendices. Note the use of lower-case Roman numerals on the preliminary pages – iii, iv, v, etc. – and Arabic numerals starting from page 1 at the beginning of Chapter 1.

## Headings

Chapter and section headings must be numbered following the pattern indicated in Section 5.5. Start each chapter on a new page and do not display chapter headings on a separate page. Sections within chapters should be continuous, not starting on a new page.

For the sake of clarity, it is best not to use more than three levels of section numbering, e.g. 1.2.3. More than this can give a staccato effect to the writing and the narrative becomes unclear. You should include at least down to the second level of section headings in the table of contents. You may omit third level headings there if it begins to make the table of contents unwieldy.

## Figures and tables

Figure captions and tables are included in the word count.

All figures and tables should be numbered, and have descriptive captions that appear below them for figures and above them for tables. They should also be listed in sequential order in separate preliminary pages near the start of your dissertation (List of figures). Numbering should be sequential within each chapter – for example, the figures in Chapter 2 would be Figures 2.1, 2.2, 2.3, etc. Where a figure or table is taken from another source, this must be acknowledged as a reference added to the title in the standard manner. Ensure that each figure or table is referred to explicitly in the text.

## 5.8 Submitting the dissertation

The dissertation must be submitted to the online TMA/EMA system by the stated submission date and time (see the study planner for due date), otherwise it will not be marked and the course will be automatically failed without permission to resubmit.

Full details and formal guidance about submitting your dissertation are given in *Information for Students Submitting Examinable Work Electronically*. You should read this document carefully when you receive it from the University's Examinations and Assessment department. A link to it will be sent to you via email approximately six weeks before the dissertation submission date. The latest version of this available to the module team can be found in the Additional resources.

### Study Note

Students studying postgraduate modules may apply to postpone submission of an EMA only if very special circumstances have affected the preparation of their project in the final few weeks immediately prior to the submission date. The criteria for such a postponement are very strict, for details see the Assessment Handbook and Section 6.4 of *Information for Students Submitting Examinable Work Electronically*.

If you are faced with a crisis which makes it difficult for you to complete the work on time and which doesn't fit the criteria for a postponement (or you uncertain whether it does), then the advice is to submit what you can and advise the T802 Examination and Award Board (EAB) through the special circumstances form. Your supervisor can advise on this and help set priorities for what to do when submitting incomplete or unfinished work.

## 6 After submission

Once you have uploaded your completed dissertation into the online TMA/EMA system you may wonder what happens to it then. Without going into too much detail, it is worth emphasising just some of the key stages of what is an exhaustive process to ensure that your dissertation is assessed in a fair and just way. It is the marking process, with its associated checks and balances, which accounts for the length of time between submission and the results becoming available.

The first stage is for each dissertation to be sent to two markers. One of the markers is the supervisor and the other is either another tutor or a member of the module team. When all the results are returned from the markers for all the dissertations, the first moderation takes place. This is where any significant discrepancies within each pair of markers are identified and action taken to resolve them. This is usually in the form of moderation where the two markers are asked to contact each other and resolve their differences. After this process second moderation takes place and at this stage dissertations are identified for third marking or review. Reviewers and third markers prepare a report for the EAB which then meets over the course of a day to decide on the final result status for each dissertation. There is one final check in the process before this becomes available to you, the author. This is whereby the actions of the EAB are ratified by Senate, the ultimate academic body in the University, for the conferring of results.

As you can see, this is quite an exhaustive process and I hope that this description gives you a feeling of confidence that each and every dissertation will be given the attention that it deserves. It may also be added that all of us associated with the T802 assessment process find it an enjoyable task and discover that we learn much from the reading of your work. This is only right and proper after all the hard work that has gone into it!